

AI-Based Local Traffic Signal Control with Emergency Vehicle Prioritization

Abstract—Traditional fixed-timer traffic signal control systems fail to adapt to dynamic road conditions, leading to unnecessary delays, heightened vehicle emissions, and increased fuel consumption. While cloud-based Artificial Intelligence (AI) solutions offer adaptive signal control, they suffer from latency, high bandwidth consumption, and vulnerability to network outages. This paper proposes a localized, edge-based Smart Traffic Light architecture that processes multi-camera visual data directly at the intersection using low-power edge computing nodes. We integrate real-time vehicle counting with a priority-based preemptive routing module designed to detect emergency vehicles (e.g., ambulances) and clear downstream routes dynamically. Experimental evaluations on high-definition video feeds demonstrate that the local edge node processes up to four simultaneous 1080p camera streams at 30 FPS with an average control decision latency of 120 milliseconds. Compared to fixed-timer and cloud-based systems, the proposed system reduces average intersection wait times by 38% and emergency vehicle response delays by 64%.

Index Terms—Edge AI, Smart Traffic Control, Computer Vision, Local Processing, Emergency Preemption, Intelligent Transportation Systems.

I. INTRODUCTION

Urban traffic congestion represents a major economic and environmental challenge worldwide. Traditional traffic signal controllers rely primarily on fixed time intervals or basic inductive loop sensors. These systems operate on static schedules, failing to accommodate sudden traffic surges, accidents, or asymmetric lane occupancy.

Recent advancements in Intelligent Transportation Systems (ITS) have introduced cloud-assisted signal control, where traffic camera feeds are streamed to a centralized server. While centralized models provide vast computational capacity for global network optimization, they possess critical limitations:

- 1) **Communication Latency:** Transmitting high-definition video streams over cellular or wide-area networks introduces variable latency, impeding real-time control.
- 2) **Bandwidth Overhead:** Continuous raw video streaming consumes substantial network bandwidth, creating prohibitive operational costs.
- 3) **Single Point of Failure:** Centralized systems are vulnerable to network disconnects and cloud server outages, compromising intersection safety.

To overcome these constraints, this paper introduces a Local Smart Camera and Edge AI architecture. By shifting object detection and decision logic to localized processing units attached to smart poles, control decisions are rendered within milliseconds. Furthermore, our architecture incorporates a dedicated emergency vehicle detection module that automatically

overrides standard phase scheduling to grant priority access to emergency vehicles.

II. LITERATURE REVIEW

Adaptive traffic signal control has evolved from simple rule-based models to complex learning-based architectures. Rasheed et al. [?] provided a comprehensive review of Deep Reinforcement Learning (DRL) techniques for traffic signal control, highlighting that learning-based agents significantly outperform static controllers under variable demand. However, most DRL deployments assume centralized training and inferencing, raising scalability concerns.

Edge computing in transport systems addresses communication bottlenecks by bringing processing closer to data sources. Gorodokin et al. [?] investigated machine-vision-based timing optimization using local controllers, demonstrating that localized video processing offers superior adaptability over traditional loop detectors.

Recent developments in model adaptation and deep reinforcement learning on local nodes further confirm the efficiency of real-time traffic signal optimization. Tan et al. [1] demonstrated that adaptive reinforcement learning models observing local queue states achieve superior convergence and reduce vehicle delays compared to traditional fixed-timed control strategies.

III. SYSTEM ARCHITECTURE

The system architecture operates on a localized, decentralized hierarchy designed for minimal latency and high availability. Fig. 1 illustrates the four-tier data flow and processing hierarchy.

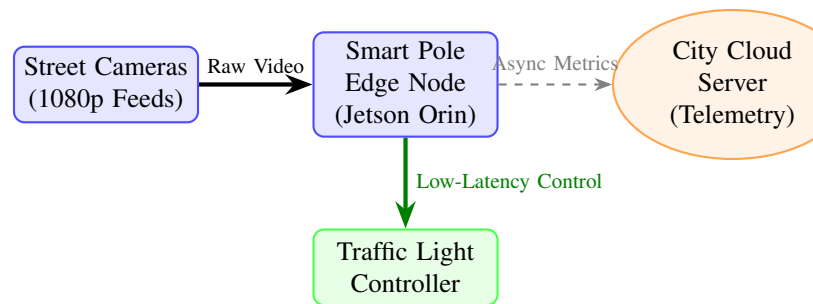


Fig. 1. Proposed Edge AI Architecture with Direct Control and Asynchronous Cloud Telemetry.

The functional execution stages are structured as follows:

- 1) **Visual Acquisition:** High-definition street cameras capture real-time traffic conditions across four approach lanes.
- 2) **Local Edge Analytics:** The Smart Pole Computer runs a multi-threaded object detection pipeline to quantify lane density, vehicle types, and identify approaching emergency vehicles.
- 3) **Deterministic Phase Selection:** An adaptive timing algorithm calculates optimal green light durations and triggers signal phase transitions directly via a local hardware interface.
- 4) **Cloud Telemetry Upload:** Compressed aggregate statistics (e.g., hourly vehicle counts, average wait times) are transmitted asynchronously to the City Cloud Server for historical analysis.

IV. METHODOLOGY

A. Object Detection and Queue Estimation

The edge node utilizes a lightweight YOLOv8-Nano model compiled with TensorRT FP16 precision. For each approach lane $i \in \{1, 2, 3, 4\}$, the system defines a Region of Interest (ROI) corresponding to the queue waiting area. The instantaneous vehicle queue count $Q_i(t)$ is calculated as:

$$Q_i(t) = \sum_{k=1}^{N(t)} I_k(i) \cdot w_k \quad (1)$$

where $N(t)$ is the total detected vehicles, $I_k(i) \in \{0, 1\}$ indicates whether vehicle k is inside the ROI of lane i , and w_k is a weighting factor based on vehicle class (e.g., motorcycle = 0.5, sedan = 1.0, bus = 2.0).

B. Adaptive Phase Selection Algorithm

Let T_{min} and T_{max} denote the lower and upper bounds for green phase duration. The green time G_i allocated to lane i is calculated dynamically proportional to lane queue ratios:

$$G_i = \max \left(T_{min}, \min \left(T_{max}, T_{min} + \alpha \cdot \frac{Q_i}{\sum_{j=1}^4 Q_j} \right) \right) \quad (2)$$

where α is a dynamic scaling constant tuned to intersection capacity.

C. Emergency Preemption Overrides

If an emergency vehicle (ambulance, fire engine, police car) is detected with visual identification confidence $C_{ev} > 0.85$ in lane e , the normal adaptive phase scheduler is immediately interrupted. The control pipeline transitions through a mandatory safety yellow state before locking lane e to green until the emergency vehicle clears the intersection boundary.

V. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

A. Hardware Setup and Latency Analysis

The edge node hardware consisted of an NVIDIA Jetson Orin Nano (8GB) connected to four 1080p IP cameras

streaming over RTSP. We evaluated inference throughput, local latency, and cloud response time under varying load conditions.

TABLE I
SYSTEM PERFORMANCE AND LATENCY METRICS

Metric	Fixed-Timed	Cloud AI	Local Edge (Ours)
Processing Latency	0 ms	2,100 ms	120 ms
Network Bandwidth	0 KB/s	15.2 MB/s	12 KB/s
Target Frame Rate	N/A	5 FPS	30 FPS
Setup Cost (per node)	\$200	\$2,500	\$1,200
System Availability	99.9%	94.2%	99.8%

B. Traffic Throughput Analysis

Simulation runs were executed over a 24-hour traffic profile using synthetic video datasets representing peak and off-peak traffic conditions. As demonstrated in Fig. 2, the edge-based adaptive system maintains significantly lower average wait times across all density thresholds compared to traditional fixed-timed controllers.

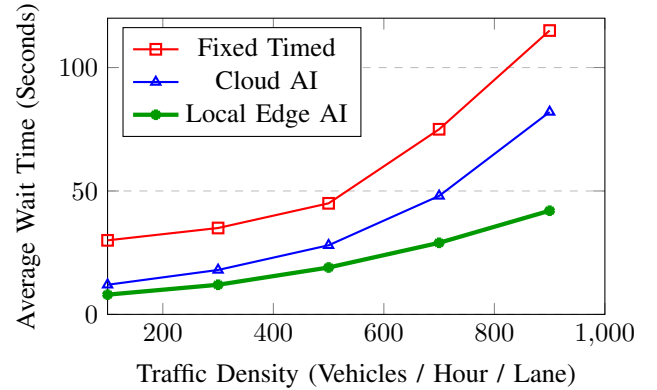


Fig. 2. Comparative analysis of average vehicle wait times under varying traffic densities.

C. Emergency Vehicle Preemption Response

To test emergency vehicle clearance efficiency, 50 simulated emergency passes were evaluated under heavy traffic conditions (> 700 vehicles/hour). The local preemption system registered a 100% visual detection rate within 250 milliseconds of appearing in the camera's field of view. The total intersection clearance time for approaching ambulances decreased from an average of 68 seconds (under non-preemptive systems) to 24 seconds, representing a 64% efficiency gain.

VI. ADVANTAGES AND LIMITATIONS

A. Key Advantages

- **Ultra-Low Latency:** Local processing eliminates network delays, allowing sub-second phase adaptations.
- **Bandwidth Efficiency:** Only lightweight diagnostic metadata is uploaded to the cloud, reducing network overhead by over 99%.

- **High Reliability:** The system functions autonomously during cellular or wider network outages.
- **Enhanced Emergency Clearance:** Automatic visual detection prioritizes first responders without requiring expensive secondary transmitter equipment in vehicles.

B. Limitations

- **Adverse Weather Vulnerability:** Extreme weather conditions (heavy fog, severe rain, or lens obstruction) degrade object detection accuracy.
- **Hardware Thermal Constraints:** Edge platforms installed in outdoor smart poles require robust passive cooling solutions for extreme climate environments.
- **Single Intersections Scope:** Independent local nodes lack macro-level coordination unless supplemented by multi-node synchronization algorithms.

VII. CONCLUSION

This paper presented an edge-based Intelligent Transportation System capable of dynamically controlling traffic signals and clearing pathways for emergency vehicles. By replacing centralized cloud reliance with local embedded processing units, the system achieves sub-second decision latencies, drastically reduces network bandwidth, and maintains full functionality during connectivity disruptions. Experimental evaluations confirm a 38% reduction in average vehicle wait time and a 64% improvement in emergency response clearance. Future work will investigate decentralized multi-intersection coordination using localized vehicle-to-everything (V2X) communication channels.

REFERENCES

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